

Supplementary Information

Landscape-scale bacterial biogeography across the Rub' al-Khali reveals recurring spatial and soil-position structure

S1 Analysis provenance

The grouped ecology, rainfall, spatial, XRF, network and encoded-function modules reported in the manuscript write machine-readable TSV and JSON results under `analysis/v3` and `analysis/xrf_audit`. Input manifests, parameter files and checksums are present only for the modules that emit them. Seeds were 20260723 for the core paired, XRF-rescue, spatial-turnover, phylogenetic and functional-redundancy analyses; 20260725 for compartment composition, depth/extraction and XRF CLR; and 20260728 for geographic prediction, spatial-resolution and distance-decay analyses. The post-hoc rainfall pulse analysis used seed 20260804 and 19,999 conditional lag rotations for each model. Its search covered 60 candidate peak times and three diversity endpoints. Sampling and peak-selection intervals used 9,999 whole-site bootstrap samples with seed 20260805. The cross-desert context analysis used seed 20260804, 9,999 depth-zone label permutations and 100 rarefactions per Atacama pit profile. The site-level climate analysis declares separate Benjamini–Hochberg families for nine diversity and 600 genus tests. Its diversity intervals used 9,999 paired-site bootstrap samples with seed 20260805. The PICRUST2 ecology analysis used seed 20260804, 9,999 whole-site permutations and 2,000 site bootstraps; its three primary pathway-profile contrasts, 600 paired pathway tests and 200 geographic pathway tests form separate declared families. The spatial and distance-decay modules use delete-one-site jackknife intervals; every pair involving the omitted site is removed in pairwise analyses. The archived-soil pH module uses dataset version EQ-PH-SHARED-v1.0.0, 999 permutations with seed 20260802, an exact input manifest, recorded software versions and output checksums under `analysis/v3/ph_shared_v1/`. The primary paired and evenness tables use the same 1,000,000-resample bootstrap implementation and seed 20260723; their estimates, intervals and unadjusted p values for the identical Shannon estimand agree. Their q values need not agree because the primary Shannon family contains three all-campaign plus 14 campaign-specific contrasts, whereas the evenness module corrects three all-campaign contrasts. Printed Shannon intervals come from `analysis/v3/results/paired_compartment_effects.tsv`; the evenness rerun must reproduce them. Multiplicity correction was applied within each declared family, not as one omnibus correction across the distinct diversity, composition, rainfall and XRF analyses. The network parameter file records its base and phase seeds. PMA mean-difference intervals used 99,999 paired-aliquot bootstrap samples and encoded-function agreement used 9,999 whole-site bootstrap samples, both with seed 20260805. A typed summary of 44 headline estimates, methods and source results is archived under `analysis/v3/headline_uncertainty/`. Table S1 distinguishes the observational units and cohorts used by each analysis.

S2 Data layers and denominator reconciliation

Table S1: Current analysis denominators. Different denominators correspond to different data products, not interchangeable estimates.

Data product	<i>n</i>	Analytical role
Observed ecology site labels	64	Numeric site labels represented in the alpha table
Primary repeated-site cohort	60 sites; 1,227 profiles	Sites 1 – 60, aggregated to 633 site-campaign-position observations
QC-passed alpha-diversity profiles	1,237	Biological profiles with at least 1,000 reads; 1,227 profiles belong to the 60-site repeated-inference cohort
Canonical all-trip feature table	1,271	Biological and control profiles; source for the 351,472-ASV feature universe
Complete field-XRF sessions	71 of 106 log entries	Trip 5 in-situ handheld measurements retained at 58 sites from a log spanning 59 sites; the log links TestID to site but not to a physical aliquot or compartment
Laboratory-XRF analytical records	725	547 records from Trips 1 – 4 plus 178 Trip 5 records; all included in the canonical elemental-axis analysis
Laboratory-XRF/alpha joins	622	Site-campaign-compartment observations on the 60 core sites
Laboratory-XRF/composition joins	621	Bray-Curtis community analysis after minimum-count filtering
Archived-soil pH/ecology joins	562 alpha; 558 composition	Quality-controlled, incomplete shared archived-sample pH cohort (EQ-PH-SHARED-v1.0.0)
Matched network cohort	170 per compartment	Same campaign-site groups in surface, shallow subsurface and root-adjacent soil; 60 sites
Resolution-matched encoded-function null	150	27 surface, 44 shallow subsurface and 79 root-adjacent profiles; 990 annotated genomes, 2,000 retained KOs and 28.5 % mean annotated-catalog fraction
PICRUSt2-shotgun overlap	125	Whole-KO-profile comparison
PICRUSt2 ecology cohort	1,227 profiles; 633 grouped profiles	Sixty core sites, 462 predicted MetaCyc pathways and 172 complete three-position site-campaign blocks
Retained CBB-marker genome records	132	RuBisCO plus phosphoribulokinase annotations; the complete upstream screened-genome manifest was not preserved, so no denominator or fraction is reported
Rainfall pulse cohort	617 site-campaign-position groups	Technical means for 196 surface, 203 shallow-subsurface and 218 root-adjacent groups across 60 sites; the paired bulk-soil sensitivity used 179 pairs; 20 distinct NASA POWER daily precipitation series
PMA pairs	9	Trip 5 treated-versus-untreated aliquots

The PMA ASV count table, 150 CoverM profiles and eggNOG-derived annotations used by the downstream analyses are enumerated with paths and checksums in the analysis manifests under **analysis/v3/**. Reproducing the upstream PMA and shotgun processing additionally requires the corresponding raw-read accessions, dereplicated MAG FASTAs and processing metadata.

The final canonical **feature-table-trips1-5.tsv** comprised 1,271 profiles and 351,472 ASVs.

The ecological dataset excluded 24 extraction-blank profiles and ten biological profiles with fewer than 1,000 reads, leaving 1,237 quality-controlled biological profiles. Of these, 1,227 profiles from sites 1 – 60 formed the repeated-inference cohort. Ten profiles from four Trip 1-only locations were retained for description but not repeated-site inference. The primary ecological analysis applied no rare-feature or contaminant filter. Hurlbert expected richness was standardized at 25,000 reads. Sixty of the 1,237 profiles, including 55 of the 1,227 core profiles, fell below that depth, leaving 1,177 full-cohort and 1,172 core expected-richness values. Shannon diversity remained available at full depth for every profile.

S2.1 Wet-laboratory and amplicon-processing record

Each field replicate was placed in a sterile tube by scooping directly with the tube or with a sterile plastic spoon. Samples were held at -20°C in an Engel MR040F portable freezer during each expedition and transferred to -80°C in the laboratory. DNA extraction used 0.9–1.2 g of soil, the QIAGEN PowerLyzer PowerSoil kit (catalogue 12855–50) or DNeasy PowerSoil Pro kit (catalogue 47014), PowerLyzer 24 homogenization, manual or QIACube Connect processing, and a $50\text{ }\mu\text{L}$ elution.

The Bakt_341F and Bakt_785R primers carried Illumina overhang adapters. Primary amplification used the QIAGEN Multiplex PCR Kit (catalogue 206143): 95°C for 15 min; 35 cycles of 94°C for 30 s, 57°C for 90 s and 72°C for 90 s; and a final 72°C extension for 10 min. Amplicons were checked on a 2 % agarose gel, indexed and cleaned following the Illumina 16S library protocol, quantified with the Qubit high-sensitivity double-stranded DNA assay, sized on an Agilent TapeStation 4200, pooled by concentration and size, and sequenced as paired 2×250 -base reads on a NovaSeq 6000.

The canonical feature table was generated with version v2.14.0 of the nf-core Ampliseq workflow. It used DADA2 and SILVA 138.2. The exact command, resource configuration and portable samplesheet are in `data/processed/amplicon/`. The retained laboratory record does not specify the PCR template input mass, exact index-kit and adapter part numbers, cleanup and indexing cycle counts, NovaSeq reagent kit, or container digest. These items remain unknown rather than inferred.

S2.2 Assay-aware control analysis

The 24 control profiles in the canonical table comprise EB1–EB18 and Negative1, Negative2, and Negative4–Negative7; all were extraction blanks. One blank was processed per extraction day, and an extraction day could include samples from several field trips. A trip value is therefore not applicable to an extraction blank. The extraction-day workbook maps EB1–EB17 to dates and 220 Trip 5 biological profiles, of which 217 occur in the canonical table. These links do not assign an EB exclusively to a trip. EB18 and the six Negative-labelled profiles lack extraction-day mappings and were retained for characterization only.

The primary screen pooled EB1–EB17 because there was only one blank per extraction batch. It used an uncorrected one-sided Fisher $p < 0.1$ and required presence in at least two blanks. Positive controls were excluded from training. The screen identified 351 of 351,472 ASVs; at the same threshold, relaxing the minimum to one blank called 12,051 ASVs. Those ASVs were removed only from the 217 biological profiles linked to the mapped Trip 5 batches; the other 1,020 ecological profiles were unchanged, and the unfiltered canonical table was retained. Removal accounted for 2.1866 % of reads across the mapped profiles, with median 0.4027 % and maximum 56.5951 % per profile. No profile fell below 25,000 reads. Shannon diversity before and after filtering had Spearman

$\rho = 0.99496$ (median change -0.00340 ; maximum absolute change 0.67805), and raw richness had $\rho = 0.99995$ (median change -11 ASVs; maximum absolute change 81). All 25 tracked headline metrics retained their scientific verdicts. Frequency modelling was not run because control DNA concentrations were unavailable.

The positive-control design was trip-specific. Trips 1 and 2 used purified-DNA ZymoBIOMICS HMW DNA Standard D6322, which can evaluate PCR, library and sequencing performance but not extraction or lysis. Trip 3 used whole-cell ZymoBIOMICS Microbial Community Standard D6300 and therefore included extraction. Trip 4 had no positive control. Trip 5 also used D6300, but its positive and negative control pair was sequenced by shotgun metagenomics rather than 16S. Seven unambiguously positive-labelled profiles in the pre-decontamination 16S table were used only for evaluation. At genus level, the two Trip 1 profiles recovered three and two of seven expected bacterial genera; the two Trip 2 profiles recovered seven and zero; and the three Trip 3 profiles recovered one, eight and eight of eight. Manufacturer genomic-DNA fractions were not interpreted as expected 16S read fractions. Under the trip-design product hypotheses, the fractions of all reads assigned to expected bacterial genera were 5.85, 12.33, 31.05, 0, 0.05, 100 and 99.99 % for `e0553.Ctrl1_2`, `e0554.Ctrl1_3`, `e0872.TMC1`, `e0874.Positive`, `e8294.FPosCtrl1`, `e8661.FPosCtrl2` and `e8665.FPosCtrl3`, respectively. Only the last two met the declared descriptive mock-dominated threshold of at least 95 %. These library-to-product assignments remain provisional; recovery does not confirm their identities. Exact mock-associated ASVs were also traced into negative controls, neighbouring indices and biological profiles, but overlap alone cannot distinguish natural occurrence, cross-contamination, index hopping or a mislabelled source library. The metadata-negative profile `e0323.Ctrl1_1.Trip1` was not promoted to a positive control from its name.

The audit follows the principle that all available processing controls should be sequenced and reported, with positive controls reserved for evaluating expected recovery rather than training a contaminant filter [7]. Three paired 16S libraries with explicit PCR or NTC labels (`e0555_PCR_Ctrl1_Trip1`, `e0875_NTC_2` and `e8667_PCRCtrl1`) and one Trip 4 workbook record already bear PCR-blank roles in the control registry. The remaining PCR-control task is a resolvable laboratory-metadata follow-up: confirm any additional reused or generic labels, their reaction or batch identifiers and the biological libraries amplified in the same batches. Until laboratory evidence confirms those links, PCR blanks remain separate from extraction blanks and cannot support a batch-linked PCR prevalence analysis. The material limits are that the Trip 4 blank covering a three-day extraction was not sequenced, affecting all or part of 23 sites; Trip 5 has no positive control in the 16S assay; and no sterile-bag inventory was recovered.

S2.3 Two XRF measurement workflows

Table S2 separates the coverage, material and analytical use of the field and laboratory workflows.

Table S2: Field and laboratory XRF are distinct measurement workflows.

Workflow	Coverage	Material	Analytical use
Field XRF	Trip 5 only; 106 log entries across 59 sites, of which 71 complete sessions at 58 sites were retained	In situ Vanta handheld sessions; TestID–site provenance only	Field-scale context and descriptive Trip 5 site-level diagnostic; not an aliquot match, interchangeability test or all-trip covariate
Laboratory XRF	All five trips; 725 records; 622 alpha and 621 composition joins	Archived collected samples measured in the laboratory	Within-campaign elemental PCA and adjusted ecology models; Trip 5 settings encoded, Trips 1 – 4 method/mode and combined-source units/LOD semantics absent

The source-specific analytical tables reproduce 547 Trips 1 – 4 records and 178 Trip 5 records. The canonical analysis includes all 725, keeps field XRF separate, excludes Trip-1-only sites 61 – 64 from repeated-campaign inference and reduces between-campaign location differences by standardizing elements within campaign. This does not remove every campaign-specific variance or method effect. Method, mode, material and diameter are encoded for Trip 5 laboratory records but absent from the consolidated Trips 1 – 4 source; concentration units, limits of detection and non-detect semantics are not machine-readable across the combined source. The Trip 5 field–laboratory comparison is a site-level diagnostic because the field log contains no physical-aliquot identifier, and it does not establish interchangeability. Trips 1 – 4 used the maximum positive analyte repeat and Trip 5 retained the last repeat. A primary-status sensitivity for the 11 affected Trip 5 analytes gave Spearman $\rho \geq 0.994$, so this bookkeeping difference did not reorder the analysis materially.

Instrument zeros were not treated as measured absence. Eleven elements detected in at least 75% of laboratory specimens (Al, Ca, Cl, Fe, K, Mg, Na, P, S, Si and Ti) were transformed as $\log(1+x)$, standardized within campaign and submitted to PCA. PC1 explained 39.2% of variance; its largest positive loadings were Ca (0.413), Mg (0.406), Na (0.370), S (0.354) and Cl (0.338), with Si negative (–0.273). PC1 correlated $r = 0.825$ with an internal evaporite-element reference and $r = 0.976$ with the globally standardized sensitivity axis.

The composition of the axis supports the descriptive term *evaporite-associated elemental axis*. In the site-fixed alpha model ($n = 622$), the Shannon coefficient was 0.00027 (95% CI [–0.040, 0.041], $p = 0.990$), so no diversity association was detected. In the primary Bray–Curtis model ($n = 621$), 38 PCoA axes retained 80.0% of variation and the elemental term had partial $R^2 = 0.00345$ ($p = 0.056$, 999 within-site permutations); retaining 95.0% of variation gave a similar $R^2 = 0.00330$ but $p = 0.044$. The estimated effect size was therefore stable across that threshold while the permutation p value was not.

Because amplicon composition is compositional, a prespecified Aitchison sensitivity reused the same 621 joins and the unchanged axis. Genera present in at least 20% of groups were ranked by mean relative abundance, and the nine combinations of 80, 200 or all 383 eligible genera with a 0.5 or 1.0 pseudocount or multiplicative zero replacement at 0.65 all supported the axis after campaign, compartment and site adjustment (partial $R^2 = 0.00311$ –0.00543, $p = 0.002$ –0.015; primary $R^2 = 0.00358$, $p = 0.010$). The primary configuration used the top 200 genera and a 0.5 pseudocount. The warranted conclusion is a small conditional composition association, identified within sites, rather than no association or an ecologically important one. A separate diagnostic

that omitted site indicators, because they force site-mean residuals to zero, found that site-mean CLR residuals remained spatially autocorrelated after campaign, compartment and elemental-axis adjustment (multivariate Moran’s $I = 0.304$, $p = 0.001$), so the axis does not fully account for the broad geographic pattern.

XRF measures total elemental composition, not soluble salts or osmotic stress. A same-matrix “salinity index” is not independent validation. Because the analysed cohort has no independent EC measurement, we use “evaporite-associated elemental axis” rather than “salinity proxy” and make no physiological salinity inference.

S3 Archived-soil pH sensitivity

The analysis used archived-soil pH dataset version EQ-PH-SHARED-v1.0.0. We measured pH in 767 specimens and retained 712 measurements with an unambiguous measurement date, a value in the physical range, recorded calibration, an electrode slope of 95 – –102 %, and a pH 10 buffer read-back of 9.9 – –10.1. 36 measurements had ambiguous dates and 19 did not meet session quality-control criteria. Missing and excluded measurements were not imputed. Coverage was incomplete and non-random, especially in campaigns 1 and 3 (Table S3).

Table S3: Coverage and quality control of the archived-soil pH measurements.

Campaign	Target specimens	Measured	Passed QC	Not measured	Excluded
1	330	157	157	173	0
2	27	23	23	4	0
3	450	192	156	258	36
4	177	175	156	2	19
5	234	220	220	14	0

The versioned SIO graph models each retained measurement as an existing specimen, a PATO acidity quality (PATO:0001842), a value typed as both an SIO measurement value (SIO:000070) and pH (SIO:001089), and an SIO measuring process (SIO:001054). Values use the pH unit (UO:0000196); process, value, quality, protocol, device and session links follow the same directions as the XRF and DNA measurement patterns. Stable IRIs depend on specimen, date and protocol rather than workbook hash. The shared version contains 712 process–quality–value sets and 29 reconstructed session groupings based on trip, date, slope and pH-10 read-back. The source workbook does not contain a session identifier. A deposited shape map enumerates every validated process, value, quality, protocol, device and session; a deposited negative fixture without the pH unit is rejected. Graph-equivalent Turtle and RDF/XML serializations are imported by the canonical knowledge graph.

The ecology join contained 702 specimens, 562 alpha-diversity groups and 558 composition groups across all 60 core sites. Table S4 reports the model results. Availability is non-random, especially within campaigns 1 and 3, so inference is bounded to this subset.

Table S4: Archived-soil pH sensitivity results for EQ-PH-SHARED-v1.0.0.

Endpoint	n	Effect size	p	Boundary
Site-fixed Shannon	562	$\beta = 0.185$	0.483	No detected diversity association
Primary top-200 CLR composition	558	partial $R^2 = 0.00274$	0.126	No detected direct association
Bray–Curtis composition, 80 % axes	558	partial $R^2 = 0.00261$	0.275	No detected direct association
Transect, same cohort before pH	60 sites	partial $R^2 = 0.358$	≤ 0.001	57 residual df; same-cohort reference
Transect, same cohort after pH	60 sites	partial $R^2 = 0.196$	≤ 0.001	57 residual df; attenuated
pH trend along transect	60 sites	partial $R^2 = 0.846$	≤ 0.001	57 residual df; archived subset

The pH maximum was the Rhizosphere group from campaign 4, site 60, based on 1 specimen. Excluding that group changed the pH transect partial R^2 from 0.846 to 0.826 and the pH-adjusted composition statistic from 0.196 to 0.187. Excluding all of site 60 gave corresponding values of 0.833 and 0.188; all refit permutation tests remained at $p \leq 0.001$. The maximum therefore influences the descriptive pH gradient but does not determine the paper’s bounded conclusion.

Across 60 sites with all 3 positions, the pH-position omnibus gave $\chi^2 = 19.757$ ($p = 5.13 \times 10^{-5}$). The Wilcoxon contrasts used 60/60, 60/60 and 59/60 nonzero/site differences for shallow-subsurface–surface, root-adjacent–surface and root-adjacent–shallow-subsurface, respectively. The differences are small; the ± 0.1 pH-10 read-back quality-control band is a session quality-control criterion, not an estimate of soil-pH analytical resolution.

S4 Evidence status and claim boundaries

Table S5 links each central result to its permissible interpretation and unresolved boundary.

Table S5: Claim ledger for the evidence-bounded manuscript.

Topic	Evidence	Permissible interpretation
Soil-position composition	170 complete campaign–site blocks; omnibus pseudo- $F = 17.28$, $p = 0.001$; pairwise-complete cohorts of 183, 182 and 191 blocks gave three CLR displacements with $p = q = 0.001$; surface-to-root-adjacent dispersion increased, but the final shallow-to-root step had $q = 0.061$	Centroid location and spread differ among operational positions; no plant-filter mechanism

Topic	Evidence	Permissible interpretation
Root proximity	Campaign-averaged, site-paired Shannon: root-adjacent-surface mean -0.130 [95 % CI $-0.321, 0.079$], $q = 0.0256$; root-adjacent-shallow mean -0.206 [$-0.398, -0.001$], $q = 0.0256$, with the bootstrap upper endpoint adjacent to zero; post-hoc normalized-evenness contrasts -0.0315 and -0.0440 , expected-richness contrasts not supported	Lower root-adjacent Shannon accompanies lower normalized evenness; no plant-filter mechanism
Geography	Centred-CLR joint linear-plus-quadratic transect model partial $R^2 = 0.4016$, $p = 0.001$; campaign-plus-block holdout $R^2 = -0.0015$; collection-order $ \rho = 0.994-1.000$; residual Moran's I depends on k	Repeatable transect structure without cross-campaign transport or geographic, or dispersal attribution; the shared pH cohort below accounts for part, but not all, of a same-cohort statistic
Archived-soil pH	702 exact specimen joins; 558 composition groups; direct primary CLR partial $R^2 = 0.00274$, $p = 0.126$; same-cohort transect partial R^2 changed from 0.358 to 0.196 after pH adjustment	pH is a measured correlate of part of the spatial pattern; no stable direct composition association, complete-survey estimate or causal mediation claim
Assembly	Top-800-ASV laboratory-XRF elemental-axis niche diagnostic: overall Mantel $r = -0.0837$ and closest-30 % pairwise $r = 0.0262$; no positive short-distance phylogenetic signal	No β NTI/RC-Bray process percentage is reported
Elemental axis	Within-campaign PC1 explains 39.2 %; adjusted Shannon $p = 0.990$; primary CLR composition partial $R^2 = 0.00358$, $p = 0.010$; all nine CLR configurations span $0.31 - -0.54$ %	Descriptive evaporite-associated elemental gradient; a small conditional composition association that does not fully account for broad geographic structure; no salinity inference
Long-term climate	Forty-nine months at each of 60 sites; all nine temperature/rain/humidity associations with Shannon, expected richness and evenness pass global correction; 111 $-$ 112 of 200 genus tests per climate summary pass a 600-test correction	Climate gradients coincide with bacterial diversity and abundance gradients; their close covariation with the transect prevents separate causal attribution
Campaign/depth	Campaign-by-position Wald $p = 0.00831$ unadjusted and $p = 0.17476$ after log-sequencing-depth adjustment; kit recorded for 858/1,237 profiles; QIACube occurs only in campaigns 4 and 5, whereas campaigns 1 $-$ 3 contain multiple recorded kits	Campaign-by-position interaction is depth-sensitive; extraction fits do not remove laboratory batch effects; no season or persistent-refugium claim

Topic	Evidence	Permissible interpretation
Rain response	Rise-and-decay model across 617 site-campaign-position groups, 60 candidate peaks and 3 endpoints; 19,999 conditional lag rotations; expected richness peaked at 2 days with NASA POWER ($p = 0.0560$) and at 1 day with Open-Meteo ($p = 0.0432$) after directional family-wise correction	Bounded evidence for a short richness association during the first several days; no causal or exact-latency claim
PMA	9 Trip 5 pairs; Hurlbert expected richness lower after treatment, no Shannon difference detected	Direct paired endpoints only; no quantitative relic fraction or survey-wide generalization
Contamination	Assay-aware screen: 17 extraction-day blanks with batch links to Trip 5 profiles; 351/351,472 candidate ASVs; 2.19% of reads removed across 217 mapped profiles; seven 16S positive profiles evaluated; 25 headline metrics rerun	All headline verdicts stable under the bounded filter; complete PCR-blank batch mapping awaits laboratory confirmation, while the absent Trip 4 positive and Trip 5 16S positive and one unsequenced Trip 4 blank bound interpretation
Conditional networks	170 matched observations per compartment; 80 genera; CLR Graphical Lasso; 200 site-cluster bootstraps; null/raw edge ratio 0.05374	Stable null-calibrated conditional associations occur within each compartment, but comparative density ordering is not stable; no interaction, hub, complexity, resilience or campaign/season claim
Encoded-functional turnover	150 analysis-ready metagenome profiles; 990 annotated genomes; 2,000 KOs; observed 0.13386 versus taxonomic 0.82272 and null median 0.08733, lower-tail $p = 1.0$, upper-tail $p = 0.001$	Lower raw encoded-functional than taxonomic turnover, but higher than the resolution-matched genome-label null; no convergence, redundancy or mechanism claim
PICRUSt2	1,227 core profiles; pathway-composition transect $R^2 = 0.234$, $p = 0.0001$; 172 complete position blocks; 125 shotgun matches with whole-KO median $\rho = 0.74$; RuBisCO $r = 0.16$, $p = 0.067$	Broad predicted potential follows geography and soil position and agrees with shotgun at whole-profile scale; no measured activity or rare-marker validation
Carbon markers	132 genome records with RuBisCO+PRK annotations; complete upstream screened-genome denominator unavailable	Candidate Calvin-cycle genomic potential, not a genome fraction or activity
Trace-gas markers	Product/KO annotation screen without catalytic, phylogenetic, synteny or direct-read confirmation	No pathway-identity or activity claim

S5 Compartment and spatial evidence

S5.1 Paired compartment composition

The primary composition analysis aggregated genus counts within campaign-site-position and retained 170 complete three-position blocks across 60 sites for the omnibus test. Genera present in at least 20% of groups were ranked by mean relative abundance; the primary configuration used

the leading 200 genera, a 0.5 pseudocount and centred-log-ratio (CLR) coordinates. Permuting position labels within blocks gave an omnibus pseudo- $F = 17.2797$ with $p = 0.001$.

Pairwise location tests used their respective pairwise-complete cohorts and sign-flipped the mean within-block CLR difference vector across sites. Standardized displacements were 0.3312 for shallow subsurface minus surface ($n = 183$ blocks), 0.4328 for root-adjacent minus surface ($n = 182$) and 0.4780 for root-adjacent minus shallow subsurface ($n = 191$); each had permutation $p = 0.001$ and Benjamini–Hochberg $q = 0.001$. The directions and location results persisted with 80, 200 or all 384 eligible genera, three zero treatments and each leave-one-campaign refit. Because the test targets the paired mean difference vector, it is exact under the paired-symmetry null even when within-position dispersion differs.

Dispersion increased monotonically from surface to shallow subsurface to root-adjacent soil. The shallow-subsurface–surface difference was 1.3776 (bootstrap 95 % CI [0.4411, 2.3748], permutation $q = 0.015$); root-adjacent–surface was 2.3866 ([1.5502, 3.2123], $q = 0.003$); and root-adjacent–shallow subsurface was 1.0090 ([0.0556, 2.0550], $q = 0.061$). The last permutation contrast does not pass $q < 0.05$, despite its bootstrap interval. The composition result describes the operational collection position and does not establish a plant selective filter.

S5.2 Alpha diversity and post-hoc normalized evenness

For all-campaign paired contrasts, profiles were averaged within site–campaign–position and then across campaigns within site and position. Among 60 sites, root-adjacent minus surface Shannon diversity had mean -0.130 (median -0.250 , bootstrap 95 % CI $[-0.321, 0.079]$, Wilcoxon $q = 0.0256$). Root-adjacent minus shallow subsurface had mean -0.206 (median -0.430 , 95 % bootstrap CI $[-0.398, -0.001]$, upper endpoint adjacent to zero, $q = 0.0256$). The endpoint’s sign is not used as evidence; the distributional claim is carried by the paired Wilcoxon result. Shallow subsurface minus surface had mean 0.075 (95 % CI $[-0.089, 0.235]$, $q = 0.189$).

A post-hoc decomposition used $H/\log(E[S_{25k}])$, where $E[S_{25k}]$ is Hurlbert expected richness at 25,000 reads. This is a normalized-evenness sensitivity, not conventional Pielou evenness. Root-adjacent minus surface normalized evenness was -0.03148 (95 % CI $[-0.04477, -0.01863]$, $q = 2.76 \times 10^{-6}$), root-adjacent minus shallow subsurface was -0.04398 ($[-0.05717, -0.03131]$, $q = 2.13 \times 10^{-7}$), and shallow subsurface minus surface was $+0.01250$ ($[0.00084, 0.02508]$, $q = 0.0121$). Campaign- and log-depth-adjusted GEE contrasts retained these directions: -0.02841 ($p = 1.67 \times 10^{-5}$), -0.04260 ($p = 3.24 \times 10^{-11}$) and $+0.01419$ ($p = 0.0125$), respectively. Expected-richness contrasts did not pass their three-test correction ($q = 0.814$, 0.0878 and 0.0675 for root-adjacent–surface, root-adjacent–shallow and shallow–surface). Hurlbert richness was unavailable for 55 core profiles below 25,000 reads, and normalized evenness was also undefined for one single-ASV profile; its GEE used 617 of 633 blocks.

S5.3 Transect structure, transport and spatial resolution

The spatial analysis retained 630 site–campaign–position groups from the 60 core sites after a 2,000-read threshold. Genus counts were CLR-transformed, centred within campaign and position, and averaged by site. With the 200 most common eligible genera, the joint linear-plus-quadratic transect trend explained partial $R^2 = 0.4016$ ($p = 0.001$, 999 site-level permutations). Linear and quadratic models using 80, 200 or all 384 eligible genera each gave $p = 0.001$, as did all five leave-one-campaign-out fits. These are influence analyses, not five replications: campaign shares of the 630 groups were 25.2 %, 2.5 %, 26.8 %, 27.9 % and 17.5 %, with the smallest campaign covering eight sites at one end of the route.

Elapsed collection time and transect position were nearly perfectly aliased: Spearman $\rho = 1.000$, 1.000, 0.9938, 1.000 and 1.000 for campaigns 1–5. The primary transport arm selected taxa within each training fold. It held out campaigns 1, 3 and 4, the campaigns with complete 60-site traversals, in turn with a contiguous ten-site block. Equal-weight $R^2 = -0.0015$, pooled $R^2 = -0.0004$, and 10 of 18 folds had positive skill; whole-site relabelling gave $p = 0.354$ and cyclic shift/reflection $p = 0.083$. Cross-campaign, cross-block transport was therefore not supported. The group-level sensitivity spanning all five campaigns was also negative ($R^2 = -0.0040$). In a sensitivity that held out a site block while retaining all campaigns on both sides, out-of-block $R^2 = 0.2526$, all six blocks had positive skill, and the same nulls gave $p = 0.002$ and 0.025. This arm does not test transport to an unseen campaign and cannot be reported as successful geographic prediction.

At ASV resolution, the same design gave partial $R^2 = 0.3737$, 0.3546 and 0.3462 for the leading 800, 2,000 and all 2,642 eligible ASVs, respectively, with $p = 0.001$ throughout. This arm used the 629-group intersection of the two caches, covering 629 of the 630 genus-reference groups; one reference group was absent from the ASV cache and two extra ASV-cache groups were dropped. The genus value above is the 630-group primary fit. Genus aggregation therefore did not create the transect result, although ASV residual Moran’s I was larger (0.149 – –0.162).

The residual diagnostic depends on the neighbour graph. This independent neighbour-count sweep used seed 20260728 and gave $p = 0.002$ at $k = 5$; the primary fixed- k analysis used seed 20260723 and gave $p = 0.001$ for the same I . Moran’s I declined from 0.1104, 0.1013, 0.0645, 0.0525 and 0.0071 at $k = 3, 4, 5, 6$ and 8 to –0.0097 at $k = 10$. Permutation $p < 0.05$ through $k = 8$ but was $p = 0.241$ at $k = 10$. Residual spatial structure is therefore a short-neighbourhood diagnostic, not a scale-independent property.

S5.4 Paired distance decay and dissimilarity components

Campaign-averaged position matrices over the same 60 sites yield 1,770 geographic pairs. Whole-site permutations were applied simultaneously to all three matrices while geographic distance remained fixed; pairwise distances were not treated as independent observations. Aitchison dissimilarity slopes per 100 km were 1.351 for surface, 1.510 for shallow subsurface and 1.180 for root-adjacent soil (each $p = 0.0001$; 9,999 permutations). The omnibus position contrast gave $p = 0.0053$. With maximum-statistic family-wise control, shallow subsurface minus root-adjacent was +0.330 per 100 km ($p_{\text{adj}} = 0.0041$), whereas surface minus root-adjacent was +0.171 ($p_{\text{adj}} = 0.182$).

After coverage standardization to 12,865 reads, mean Sørensen dissimilarity was 0.401, 0.406 and 0.395 in surface, shallow-subsurface and root-adjacent soil. Simpson replacement accounted for 0.286, 0.280 and 0.295, or 69 – –75 % of Sørensen dissimilarity, and both replacement and nestedness components increased with distance under the same permutation maps. Their slopes per 100 km were, respectively, 0.0075 versus 0.0113 in surface soil, 0.0097 versus 0.0165 in shallow-subsurface soil, and 0.0117 versus 0.0055 in root-adjacent soil. Replacement therefore dominated mean dissimilarity in all three positions, but not the distance-related increase in either bulk-soil position. The unstandardized Bray–Curtis abundance-gradient component correlated with the absolute log library-size ratio at $r = 0.76$ –0.83 and becomes identically zero at common depth, so it supports no ecological claim.

S5.5 Conditional association diagnostic

Thresholded Spearman networks contained 90.6 – –99.9 % positive edges, a warning for compositional and indirect correlation artifacts, and were not interpreted. For the conditional association diagnostic, sequencing replicates were summed within campaign–site–compartment, aggregate read

depth was required to be at least 2,000, and the analysis intersected the three compartments to 170 observations each across 60 sites. Campaign contributions were 42, 2, 53, 56 and 17 observations per compartment. The 80 most abundant genera present in at least 20 % of every compartment were CLR-transformed (pseudocount 0.5), centered within campaign and standardized. A common Graphical-Lasso penalty $\alpha = 0.20$ was selected against a permuted-taxon null; its combined null/raw edge ratio was 0.05374. Stable edges required at least 0.80 selection and 0.90 sign consistency in 200 paired site-cluster bootstraps and null selection frequency at most 0.05. Table S6 reports the resulting stable-edge diagnostics.

Table S6: Stable null-calibrated conditional associations by operational soil position. Expected false fractions are calibration diagnostics, not formal false-discovery-rate estimates.

Compartment	Stable edges	Stable density	Positive fraction	Expected false fraction
Surface	180	0.05696	0.6278	0.0096
Shallow subsurface	221	0.06994	0.5747	0.0087
Root-adjacent	237	0.07500	0.5865	0.0104

Stable null-calibrated conditional associations were therefore recoverable in every compartment. The mean expected false fractions among stable edges are calibration diagnostics, not formal false-discovery-rate estimates. Comparative density ordering was not stable: 95 % site-cluster bootstrap intervals were $[-0.0142, 0.0089]$ for surface minus shallow subsurface, $[-0.0161, 0.0070]$ for surface minus root-adjacent and $[-0.0155, 0.0104]$ for shallow subsurface minus root-adjacent. Density directions persisted in only 0.78, 0.67 and 0.67 of nine taxon-count/penalty configurations. Campaign networks were not estimated because no common campaign comparison met the predeclared 80-observation rule. The retained edges are descriptive regularized conditional associations, not interactions.

S6 Phylogenetic diagnostic

We tested whether phylogenetic distance was positively associated with niche difference, a prerequisite for interpreting β NTI/RC-Bray categories, using the same top-800 abundant-ASV set. Each ASV's niche summary was its abundance-weighted mean on the canonical laboratory-XRF elemental axis. Among the closest 30 % of ASV pairs the association between phylogenetic distance and niche difference was negligible ($r = 0.0262$), and over all pairs it ran in the opposite direction to the requirement (Mantel $r = -0.0837$, $p = 0.002$): the closest distance class had a larger mean niche difference than the farthest (0.808 versus 0.515). The prerequisite was not met, so we do not interpret β NTI/RC-Bray assembly-process categories for these data.

S7 Climate, campaign, rainfall and PMA evidence

A site-clustered Gaussian GEE gave a campaign-by-position interaction for Shannon diversity before sequencing-depth adjustment (Wald $p = 0.00831$) but not after adjustment for log read depth ($p = 0.17476$). Campaign-specific shallow-subsurface-surface paired means were +0.434 in Trip 1 (bootstrap 95 % CI 0.241–0.633), +0.216 in Trip 3 ($[-0.003, 0.433]$), -0.212 in Trip 4 ($[-0.516, 0.066]$) and +0.127 in Trip 5 ($[-0.286, 0.526]$); Trip 2 lacked enough complete pairs. The depth-adjusted root-adjacent-shallow contrast remained supported at -0.273 (95 % CI $[-0.424, -0.122]$), while

shallow-surface was direction-only at $+0.103$ ($[-0.012, 0.217]$, $p = 0.0791$). Across the six pre-specified extraction models, the root-adjacent-shallow contrast excluded zero in every fit. The root-adjacent-surface direction was stable, but its interval excluded zero in only three fits (additive unadjusted, additive depth-adjusted and protocol-matched pairs), and not in the recorded-kit, complete-case or kit-varying-campaign fits.

Extraction kit was recorded for 858 of 1,237 profiles and was partly confounded with campaign. QIACube occurred only in campaigns 4 and 5, which contained no other recorded kit, so its indicator was rank-deficient given campaign among complete cases. Campaigns 1 – 3 contained more than one recorded kit. The log-depth and recorded-kit fits are therefore sensitivities rather than adjustment for all laboratory batch effects. The campaign-interaction dependence is shown for Shannon diversity and is not claimed for alpha diversity generally. Season, campaign, date and processing context remain inseparable.

S7.1 Long-term climate and collection-weather associations

The field-weather ledger contained 247 quality-controlled primary-site air temperatures ($20.0 - -49.4^{\circ}\text{C}$), 177 atmospheric-pressure values ($894 - -998$ mbar) and 239 relative-humidity values ($9.5 - -62.5\%$). We first averaged the ecological endpoints within site and campaign. Raw pressure was correlated with Shannon diversity, expected richness and normalized evenness ($\rho = -0.570$, -0.591 and -0.349). These associations did not pass the nine-test correction after campaign and a quadratic transect coordinate were included with site-clustered standard errors (all adjusted $q \geq 0.527$). Neither field temperature nor field humidity had a corrected adjusted association. These records are air conditions observed during a site visit; they are not replicated soil-temperature or moisture measurements.

The long-term analysis used 49 monthly Open-Meteo observations at every core site, from January 2022 through January 2026 [25]. The mean ranges were $26.38 - -30.30^{\circ}\text{C}$ for temperature, $1.31 - -7.47$ mm per month for rain and $24.27 - -35.40\%$ for relative humidity. Table S7 reports the nine site-level diversity associations. All passed one global Benjamini-Hochberg correction. Expected richness used the available standardized values within each site; all 60 sites remained represented.

Table S7: Site-level Spearman associations between 49-month climate summaries and bacterial diversity. Brackets give 95 % paired-site percentile bootstrap intervals from 9,999 samples. The correction covers all nine tests.

Climate summary	Diversity measure	ρ [95 % interval]	q
Temperature	Shannon	-0.711 $[-0.833, -0.520]$	6.02×10^{-10}
	Expected richness	-0.795 $[-0.887, -0.639]$	3.12×10^{-13}
	Evenness	-0.584 $[-0.753, -0.359]$	1.75×10^{-6}
Rain	Shannon	-0.470 $[-0.671, -0.211]$	1.67×10^{-4}
	Expected richness	-0.495 $[-0.702, -0.226]$	7.51×10^{-5}
	Evenness	-0.298 $[-0.532, -0.023]$	0.0206
Relative humidity	Shannon	-0.701 $[-0.834, -0.490]$	1.04×10^{-9}
	Expected richness	-0.770 $[-0.877, -0.593]$	2.87×10^{-12}
	Evenness	-0.522 $[-0.713, -0.266]$	2.89×10^{-5}

For genus abundance, the analysis selected the 200 genera with the largest mean relative abundance among those present in at least 20 % of profiles. It applied a 0.5 pseudocount, calculated profile-level CLR values and averaged them within site. Across one global family of 600 tests, 112

genera were associated with temperature, 112 with rain and 111 with humidity. The strongest positive temperature associations included *Halalkalibacter* ($\rho = 0.845$) and *Bacillus* ($\rho = 0.833$); the strongest negative associations included *Cellulomonas* and *Marmoricola* (both $\rho = -0.690$). *Truepera* increased with the rain gradient ($\rho = 0.700$), whereas *Massilia* decreased ($\rho = -0.635$).

The climate summaries were not independent of geography. Their correlations with the transect coordinate were $\rho = 0.982$ for temperature, 0.551 for rain and 0.919 for humidity. Temperature and humidity were also correlated with one another ($\rho = 0.894$), as were rain and humidity ($\rho = 0.630$). The site-level results therefore describe environmental gradients that coincide with the bacterial geography. They do not estimate separate causal effects of each climate variable. Exact site summaries, correlations, genus rankings, field weather models, software versions and checksums are archived under `analysis/v3/environment_associations/`.

S7.2 Fitted short-term rainfall pulse

The primary pulse model used the unfiltered community table and technical means for every available site, expedition and soil position. This gave 617 analysis units: 196 surface, 203 shallow-subsurface and 218 root-adjacent groups. Daily precipitation came from the versioned NASA POWER PRECTOTCORR series [17]; the 60 sites resolved to 20 distinct daily rainfall series. Open-Meteo, which resolved 58 series, provided an independent product sensitivity. We excluded rain on the sampling date because collection times were unavailable.

For each candidate peak from 0.5 to 30 days, the model weighted the preceding 60 complete days with the rise-and-decay curve defined in the main Methods. Expected richness at 25,000 reads, Shannon diversity and normalized evenness formed one search family. The nuisance model included site-by-position baselines, expedition effects, and expedition-specific linear longitude and collection-day trends. The search statistic was the largest positive classical t value across all 60 peaks and three endpoints.

We calibrated this complete search with 19,999 conditional rotations (seed 20260804). Each rotation cyclically moved the 60-day lag axis by one independently sampled amount within each expedition. This preserves the rain amount, rarity, storm clustering and spatial field observed in every expedition, while breaking its timing relative to collection. It therefore does not require rain to recur across expeditions. A plus-one directional p value tested the positive pulse hypothesis. A second maximum used absolute t values for a two-sided familywise sensitivity. Site-clustered intervals described the selected effect, and an exact 120-member cyclic and reflected route orbit tested sensitivity to the spatial rain field. We also drew 9,999 multinomial whole-site bootstrap samples (seed 20260805), keeping every observation from one site together. Fixed-curve draws quantified effect and partial- R^2 variation after the fitted nuisance adjustment. Separate draws reselected the peak for the already selected richness endpoint, giving a peak-selection interval rather than treating the observed peak as fixed. A full-search diagnostic also reselected the endpoint. These intervals are conditional on the fitted nuisance model and one rainfall product; product and route sensitivity remain separate questions.

With NASA POWER, expected richness gave the largest statistic. The curve selected a two-day peak and an estimate of 179.6 expected taxa per millimetre at the kernel peak (site-clustered 95 % CI 92.1 – 267.1). Rain accounted for 3.64 % of the variation remaining after the nuisance model. The complete directional search gave $p = 0.0560$, the two-sided sensitivity gave $p = 0.0787$, and the spatial route orbit gave $p = 0.0833$ in the positive direction. The fitted curve fell to half its peak by 5.36 complete days and to 10 % by 9.78 days. The bootstrap peak-selection interval was 1.5 – 4.5 days; the half-peak and 10 % decay-time intervals were 4.02 – 12.05 and 7.33 – 22.00 days. Open-Meteo selected a one-day peak (directional familywise $p = 0.0432$; two-sided $p = 0.0487$),

with a 0.5 – –6.5-day peak-selection interval. Its corresponding decay times were 2.68 and 4.89 days, with much wider intervals of 1.34 – –17.41 and 2.44 – –31.78 days. The products therefore agree on an early association but do not identify an exact delay or decay time.

When the endpoint was also reselected in each site bootstrap, richness was chosen in 73.7 % of NASA POWER samples and 79.7 % of Open-Meteo samples. Shannon was chosen in 24.7 % and 6.2 %, and evenness in 1.6 % and 14.1 %, respectively. This is selection uncertainty, not a second hypothesis test.

Table S8 separates effect, explained-variation and timing uncertainty for the two rainfall products.

Table S8: Effect-size and timing uncertainty for the two rainfall products. Effect intervals use site-clustered sandwich covariance. Partial- R^2 intervals use the fixed-curve site bootstrap. Peak intervals reselect the richness peak in every site bootstrap. The final value gives the frequency with which richness was selected when both endpoint and peak were reselected.

Product	Effect per mm [95 % CI]	Partial R^2 [95 % interval]	Peak, d [95 % interval]	Richness selected
NASA POWER	179.6 [92.1,267.1]	0.0364 [0.0060,0.0787]	2.0 [1.5,4.5]	73.7 %
Open-Meteo	327.5 [141.1,514.0]	0.0384 [0.0057,0.0899]	1.0 [0.5,6.5]	79.7 %

A flexible shape sensitivity fitted 7 disjoint lag intervals together in each rainfall product. For the NASA POWER series, the largest positive estimate occurred at 3–4 days (134.8 expected taxa per millimetre, site-clustered SE 32.4). The Open-Meteo series also placed its largest positive estimate at 3–4 days (77.8, SE 31.7). These jointly fitted coefficients are descriptive shape checks, not separately corrected tests. They support the first several days as the interpretable window without fixing an exact delay. Future rainfall produced no comparable richness pattern in the primary product: its largest richness statistic was 0.055. The largest future statistic instead occurred for evenness at 19.5 days ($t = 2.13$); this negative control was descriptive and did not enter the primary correction.

The selected richness estimate was positive in surface (176.4), shallow-subsurface (228.0) and root-adjacent soil (136.8 expected taxa per millimetre). Contaminant filtering barely changed the estimate (179.4), even though its corrected directional probability crossed the conventional threshold ($p = 0.0485$). We therefore do not use filtering to strengthen the effect. pH adjustment retained the early positive pattern, while a complete paired bulk-soil cohort selected Shannon diversity at two days with weaker evidence. More flexible route terms retained the positive direction but usually weakened the correction (Table S9).

Table S9: Rainfall pulse-response analyses. p_+ is corrected over all peak times and endpoints for the positive pulse hypothesis; $p_{\pm}$ is the two-sided familywise sensitivity. The route-orbit probability tests the spatial rainfall field. Filtering changes the primary probability but not the effect estimate.

Analysis	n	Selected endpoint	Peak (d)	p_+	$p_{\pm}$	Route p_+
NASA POWER, unfiltered	617	Expected richness	2.0	0.0560	0.0787	0.0833
Open-Meteo, unfiltered	617	Expected richness	1.0	0.0432	0.0487	0.0333
Control-filtered	617	Expected richness	2.0	0.0485	0.0711	0.0833
pH-adjusted cohort	547	Expected richness	2.5	0.0480	0.2608	0.0750
Paired bulk soil	179	Shannon diversity	2.0	0.0804	0.2700	0.2417
Quadratic longitude	617	Expected richness	4.5	0.0904	0.4650	0.2333
Quadratic longitude and day	617	Expected richness	2.0	0.0909	0.6763	0.2250
Cubic longitude	617	Expected richness	2.5	0.0478	0.3894	0.1500

For the four complete days before Trip 1 samples, we also compared five gridded products: CHIRPS v2.0, CMORPH V1.0, GPM IMERG Final V07B, NASA POWER and Open-Meteo [8, 23, 10]. Every pair was positively correlated across the 60 sites (Pearson $r = 0.501$ – 0.917 ; Spearman $\rho = 0.554$ – 0.881). This fixed-event comparison supports the spatial exposure pattern, but only NASA POWER and Open-Meteo cover the complete fitted-window sensitivity. Source filenames, official download addresses, byte sizes and checksums are recorded under `data/processed/climate/rain_event_product_sources.tsv`.

Complete cohorts, exposures, scans, nulls, position effects, negative controls, software versions and checksums are archived under `analysis/v3/rain_pulse_response/` and `analysis/v3/rain_pulse_sensitivities/`. Figure S1 shows the primary fitted curve and its calibrated search.

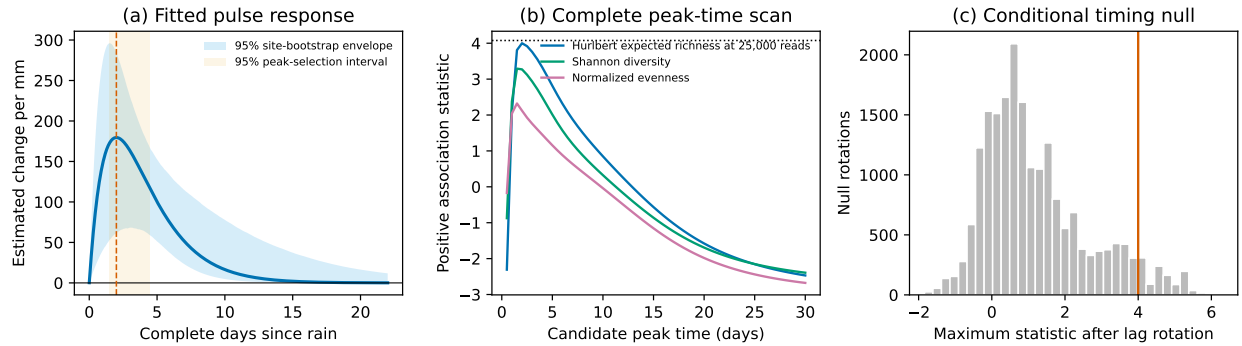


Figure S1: Fitted antecedent-rain pulse. (a) Estimated change in expected richness after one millimetre of rain under the selected NASA POWER pulse kernel. The blue band is the 95 % site-bootstrap curve envelope, the orange band is the 95 % peak-selection interval, and the dashed line marks the two-day fitted peak. (b) Positive-association statistics across all candidate peak times for expected richness, Shannon diversity and normalized evenness; the dotted line is the 95th percentile of the complete-search null. (c) Conditional null distribution of the largest positive statistic across all peak times and endpoints under 19,999 within-expedition lag rotations; the red line marks the observed maximum. The familywise directional probability was 0.056. The curve estimates an association, not a causal response or exact biological latency.

S7.3 Paired PMA endpoints

For the PMA pairs, Hurlbert expected richness at the minimum common depth of 123,897 reads was 771.8 untreated versus 608.5 treated (exact paired Wilcoxon $p = 0.00781$; eight of nine pairs lower). The mean treated-minus-untreated difference was -163.3 (95 % paired-aliquot bootstrap interval -274.0 to -69.8). Shannon was 4.19 versus 4.32; its mean difference was 0.130 (95 % interval -0.154 to 0.467 ; $p = 0.652$), so no difference was detected. Both intervals used 99,999 resamples of the nine matched pairs with seed 20260805. The manuscript reports the paired treatment response only. The surviving protocol record does not preserve the PMAxx concentration, dark-incubation time, photoactivation duration or photoactivation device, preventing exact replication of the wet-laboratory treatment. PMA-seq is qualitative in complex communities and is affected by matrix, biomass and taxonomic composition [22]. The staged ASV count table supports re-execution of these endpoints; reproducing it from reads requires separate public raw-read accessions.

S8 Predicted pathway structure and shotgun encoded function

S8.1 PICRUSt2 ecology analysis

PICRUSt2 v2.4.1 predictions matched all 1,227 profiles from the 60 core sites. The unstratified input contained 462 MetaCyc pathways [3]. Sequencing replicates were summed within campaign, site and soil position, yielding 633 grouped profiles. Of the 462 pathways, 416 occurred in at least 20 % of grouped profiles. The primary analysis selected the 200 with the largest mean relative abundance and applied a 0.5 pseudocount before CLR transformation. Analyses of 100 pathways, all 416 prevalent pathways and a 1.0 pseudocount tested the feature-set and zero-treatment choices.

For geography, campaign-by-position means were removed before profiles were averaged within each of the 60 sites. A pre-specified linear and quadratic transect model accounted for 23.42 % of predicted pathway composition (95 % delete-one-site jackknife interval, 15.46–31.39 %; $p = 0.0001$; 9,999 whole-site permutations). Across pathway-set, pseudocount and leave-one-campaign analyses, the explained share ranged from 21.39 to 36.26 %, and every permutation test gave $p = 0.0001$.

For soil position, 172 campaign-by-site blocks contained surface, shallow-subsurface and root-adjacent profiles. Each complete block was centred on its own three-position mean and then averaged within site, making the 60 sites the independent units. The omnibus position statistic was pseudo- $F = 6.37$ ($p = 0.0001$). Standardized paired displacements were 0.453 for root-adjacent minus shallow subsurface ($q = 0.00015$), 0.429 for root-adjacent minus surface ($q = 0.00015$), and 0.330 for shallow subsurface minus surface ($q = 0.0194$). The omnibus result and both root-adjacent contrasts passed every pathway-set, pseudocount and leave-one-campaign test. The shallow-subsurface–surface contrast did not pass when Trip 3 was omitted ($p = 0.136$), so it is the weaker pairwise result.

Pathway-wise sign-flip tests retained the same site-level paired design. One global correction covered 600 pathway-by-position tests: 121 root-adjacent–shallow, 73 root-adjacent–surface and 76 shallow–surface associations passed $q < 0.05$. The supported predictions included amino-acid and cofactor biosynthesis, carbohydrate degradation and cell-envelope pathways. A separate family of 200 pathway–transect correlations retained 92 pathways. These follow-up tests identify predicted pathways for direct measurement; because they are derived from the ASVs, they are not independent evidence of metabolic activity.

Several supported predictions motivate more specific follow-up experiments. Along the west–east route, cis-vaccenate biosynthesis, gondoate biosynthesis and saturated fatty-acid elongation each had $\rho = 0.610$ – 0.616 ($q \leq 3.24 \times 10^{-6}$), whereas palmitate biosynthesis had $\rho = -0.527$ ($q =$

1.18×10^{-4}). Root-adjacent soil exceeded both bulk-soil positions for predicted glucose and xylose degradation, fructuronate and 4-aminobutanoate degradation, polyamine biosynthesis and thiamin diphosphate biosynthesis (all $q \leq 0.0162$). These are pathway predictions, not measurements of membrane adaptation, plant-derived substrate use or metabolite exchange. The selected MetaCyc set did not contain a dedicated UV-repair pathway. Testing UV protection or DNA repair therefore requires a pre-specified gene-level analysis rather than an interpretation added to this pathway screen. The complete 600 position tests and 200 geographic tests are in [analysis/v3/picrust2_ecology/](#).

Weighted NSTI had median 0.1648 across the 1,227 profiles (range 0.0053 – -0.7819). The median was 0.2011 in shallow-subsurface, 0.1681 in surface and 0.1340 in root-adjacent profiles. Lower NSTI means that the contributing ASVs are closer to the reference genomes used for prediction. Exact grouped cohorts, primary and sensitivity tests, pathway-level results, NSTI summaries, seeds and checksums are archived under [analysis/v3/picrust2_ecology/](#).

S8.2 Resolution-matched shotgun encoded-function test

Table S10 reports the resolution-matched functional turnover test and its intact-genome-label null.

Table S10: Resolution-matched encoded-function null. “Encoded” is the median Bray–Curtis dissimilarity of abundance-weighted KO profiles; the null reassigns intact genome annotation profiles among fixed genome-abundance vectors (999 permutations). p_L and p_U are the plus-one-corrected lower- and upper-tail values.

Annotation basis	Group	n	Taxonomic	Encoded	Encoded/taxonomic	Null median [95 % interval]	p_L	p_U
KO copy count	All	150	0.82272	0.13386	0.16271	0.08733 [0.08459, 0.09067]	1.000	0.001
	Shallow subsurface	44	0.80929	0.13270	0.16397	0.08400 [0.08029, 0.08838]	1.000	0.001
	Root-adjacent	79	0.81810	0.13348	0.16316	0.08889 [0.08567, 0.09269]	1.000	0.001
	Surface	27	0.75209	0.11254	0.14964	0.08138 [0.07595, 0.08765]	1.000	0.001
Within-genome relative	All	150	0.82272	0.13659	0.16602	0.08882 [0.08592, 0.09315]	1.000	0.001
	Shallow subsurface	44	0.80929	0.13615	0.16824	0.08544 [0.08165, 0.09040]	1.000	0.001
	Root-adjacent	79	0.81810	0.13418	0.16402	0.09041 [0.08702, 0.09523]	1.000	0.001
	Surface	27	0.75209	0.11573	0.15388	0.08284 [0.07758, 0.09023]	1.000	0.001

The staged, checksum-tracked CoverM and eggNOG-derived inputs support re-execution of the following downstream test; they do not support regeneration from raw reads without the separate MAG catalogue and upstream metadata. The observed encoded-functional dissimilarity was greater, not smaller, than the complete 95 % null interval under both annotation bases and in every compartment. Accordingly, encoded-functional turnover was lower than taxonomic turnover in absolute terms, but higher than the resolution-matched genome-label null. Functional convergence or redundancy beyond feature resolution is not supported. Observed turnover exceeded every null replicate ($p_U = 0.001$), but this test does not identify a mechanism. The null preserves the genome-abundance matrix, every intact annotation profile and KO prevalence; it is not a metabolic-network null. Only 0.28465 of the CoverM total was assigned to the annotated 990-genome set on average. The result therefore concerns genome-resolved encoded potential in a minority fraction, not direct-read function, expression, activity or flux.

A top- K sensitivity retained 1,000, 2,000 or 5,000 KOs. Observed median KO-copy Bray–Curtis dissimilarity was 0.1005, 0.1339 and 0.1721, compared with label-null medians of 0.0670, 0.0873 and 0.1174, respectively. Observed turnover exceeded the null upper interval in all three configurations ($p_U = 0.001$ throughout), so the conclusion does not depend on the primary $K = 2,000$ choice.

Across 125 samples and 3,471 shared KOs, the PICRUSt2-shotgun whole-profile Spearman correlation had median 0.743 (95 % whole-site bootstrap interval 0.731 – -0.751) and community-

mean 0.806 (0.793 – –0.812). The 9,999 bootstrap samples kept all matched profiles from the same field site together and used seed 20260805. Rare-marker agreement is much weaker and not distinguishable from zero (RuBisCO $r = 0.16$, two-sided $p = 0.067$), so the whole-profile association is not treated as rare-marker validation. These predictions used separately executed Trips 1 – –4 and Trip 5 input sets containing 330,830 ASVs, rather than the 351,472-ASV combined canonical table used for the primary ecological analyses.

The staged positive-only metabolic-marker table contains 132 genome records annotated with both RuBisCO and phosphoribulokinase. The complete upstream screened-genome manifest was not preserved in the local analysis bundle, so no denominator or genome fraction is reported. This supports candidate Calvin-cycle genomic potential in the recovered output. The supplied pathway screen contains no positive calls for rTCA, Wood–Ljungdahl or the 3-HP cycle. It contains 33 single-marker hits for the hydroxybutyrate cycle; the latter marker is not pathway-specific, and zero calls are not evidence of biological absence. The result does not exclude phototrophy, show carbon fixation or identify an energy source. Hydrogenase and CO-dehydrogenase product labels are not sufficient to establish high-affinity atmospheric oxidation; *mcrA* and *nifH* non-detections are bounded by recovery and annotation sensitivity. No profile-HMM, catalytic-motif, phylogenetic, synteny or direct-read confirmation was included, so no trace-gas pathway or activity claim is made.

S9 Cross-desert and regional context

S9.1 Quantitative comparison with public Atacama surveys

We compared effects rather than merging feature tables because the surveys used different 16S regions, reference databases and sampling designs. The Atacama aridity-gradient input is an archived site-level table derived from QIITA study 10360. The upstream V4 OTU profiles were rarefied 100 times to 12,000 reads and technical replicates were averaged within site. The replayed analysis retained 16 sites with complete soil relative humidity and spatial information. Soil relative humidity was positively associated with Shannon diversity ($\rho = 0.8904$, $p = 3.80 \times 10^{-6}$). The association remained after ranked humidity and diversity were adjusted for latitude, longitude and elevation (partial $\rho = 0.6753$, $p = 0.00409$). Rarefied richness gave the same direction (raw $\rho = 0.8712$; partial $\rho = 0.6445$).

For the Atacama pit, all 62 reprocessed intracellular-DNA profiles exceeded 8,000 reads. Replicates represented 23 sampled depths from 2.5 to 420 cm. We combined replicate counts within depth, retained the 200 leading genera and tested three zones in CLR coordinates. Composition differed among zones (pseudo- $F = 1.463$, $p = 0.0014$, 9,999 label permutations). After 100 rarefactions per profile and averaging within depth, richness increased only weakly with depth ($\rho = 0.381$, $p = 0.0725$), and Shannon diversity had no monotonic trend ($\rho = 0.120$, $p = 0.587$). The compositional result supports depth as a recurring source of habitat structure; the diversity direction is not portable from one depth design to another.

These comparisons do not identify a common cause. Soil relative humidity in the Atacama and a 49-month atmospheric humidity mean in the Empty Quarter are different exposures. The Atacama pit is one vertical profile sampled to 4.2 m, whereas Empty Quarter positions span only 0 – –10 cm and include loose root-adjacent soil. Exact inputs, parameters, software versions and checksums are recorded under `analysis/v3/cross_desert_context/`. The archived Atacama-gradient table reproduces the site-level tests, but regeneration of that table from the original QIITA BIOM file remains an upstream step.

S9.2 Design boundaries in related studies

Table S11 places the survey against regional observational and experimental precedents while retaining their design boundaries.

Table S11: Studies that define the comparison and evidential boundary.

Study	Design	Relevance to this manuscript
McKay et al. 2016, “An Unusual Inverted Saline Microbial Mat Community in an Interdune Sabkha in the Rub’ al Khali” [15]	One groundwater-fed Liwa sabkha, UAE; salt mat microbiology and geochemistry	Direct Rub’ al-Khali precedent, but a specialized saline mat rather than open Saudi landscape soils
Yasir et al. 2015, “Composition of soil microbiome along elevation gradients in southwestern highlands of Saudi Arabia” [24]	Ten soils across four southwestern highland elevations; 16S V6 pyrosequencing and soil chemistry	Early Saudi spatial soil-microbiome precedent; a small highland gradient rather than a repeated hyperarid transect
Aslam et al. 2016, “Soil compartment is a major determinant of the impact of simulated rainfall on desert microbiota” [1]	In-situ watering of Saudi desert crust, root-attached and broader root-zone soil	Direct experimental compartment/rainfall comparator; shows why our five observational dates cannot substitute for a manipulated response trajectory
Eida et al. 2018, “Desert plant bacteria reveal host influence and beneficial plant growth properties” [5]	Barren soil, rhizosphere and root endosphere of identified hosts in Jizan and Al Wahbah	Defines the sampling and host metadata needed for rhizosphere/endosphere and host-filter claims
Khan and Khan 2020, “Microbial communities and their predictive functional profiles in the arid soil of Saudi Arabia” [12]	Abha, Hafar Al Batin and Muzahmiya; 16S and PICRUSt; includes rhizosphere	Direct Saudi arid-soil precedent; prevents a “first Saudi survey” claim
Maurice et al. 2023, “Fertility islands, keys to the establishment of plant and microbial diversity in a highly alkaline hot desert” [14]	AlUla vegetation patches, soils, geochemistry and microbial diversity	Saudi fertility-island and plant-associated geochemistry comparator; unlike operationally defined root-adjacent soil
Maurice et al. 2024, “Networking the desert plant microbiome” [13]	Roots and rhizospheres of six identified plant species in AlUla, Saudi Arabia	Shows the host and root metadata needed for a plant-selective-filter mechanism

Study	Design	Relevance to this manuscript
Jalal et al. 2022, Saudi coastal halophyte compartments [11]	Southern Corniche halophyte-associated soil, rhizosphere, root and leaf endosphere samples	Saudi saline-soil precedent; a specialized host system that does not validate salinity physiology in our XRF gradient
Mousa et al. 2024, “High-throughput sequencing reveals the structure and metabolic resilience of desert microbiome confronting climate change” [16]	Rhizosphere and endosphere sequencing of native Arabian desert plants in Dubai and Al Ain, UAE	UAE/Arabian plant-desert comparator; distinct from the open-soil landscape design used here
Pan et al. 2022, “Aridity Threshold Induces Abrupt Change of Soil Abundant and Rare Bacterial Biogeography” [18]	Chinese dryland gradient spanning aridity threshold; abundant/rare assembly	Hypothesis-level comparison; our one-dimensional transect cannot identify an aridity response
Bay et al. 2020, Negev soil biogeography [2]	99 soils across as much as 189 km; fine-resolution community turnover and distance decay	Independent hot-desert support for spatial turnover; primer, distance metric and landscape differ, so coefficients are not directly comparable
Ronca et al. 2015, Namib dune/interdune transects [20]	Surface soils across repeated dune and interdune habitats	Shows habitat-specific turnover in another sand desert; it does not provide the repeated, three-position design used here
Pombubpa et al. 2020, Mojave biocrust survey [19]	48 surface and subsurface profiles across four sites and crust types	Independent support for depth-related composition; biocrust and open-sand positions are different habitats
Demergasso et al. 2023, “Hyperarid soil microbial community response to simulated rainfall” [4]	Controlled, replicated 30-day rainfall treatment	Shows the trajectory and controls needed to establish bloom/relaxation
Horstmann et al. 2024, “Persistent microbial communities in hyperarid subsurface habitats of the Atacama Desert” [9]	Depth profile to 4.2 m; intracellular DNA	Appropriate subsurface comparison; our 5 – –10 cm samples do not show a deep refugium or activity
El Moatasssem et al. 2025, “Soil salinity hazard monitoring with portable X-ray fluorescence spectrometry and electrical conductivity meters” [6]	pXRF models calibrated against EC	Establishes the independent validation required before calling the XRF axis salinity

Study	Design	Relevance to this manuscript
Tribbia et al. 2026, “Trace gas oxidation supports sub-surface microbial communities across Namib Desert fog and aridity gradients” [21]	Gas uptake, wetting, ATP, respiration, abundance and geochemistry	Defines the direct activity evidence missing from our marker inventory

This study is a large, repeated landscape survey of open sand, shallow subsurface and root-adjacent Saudi Rub’ al-Khali soils, joined to distinct field/laboratory XRF products and an analysis-ready shotgun subset. It is not the first Saudi arid-soil study, first Rub’ al-Khali microbial study, proof of a seasonal bloom, or evidence of active atmospheric chemosynthesis.

S10 Uncertainty estimates for headline results

Table S12 lists the point estimates that carry the main ecological interpretation. Brackets are 95 % sampling intervals unless they are identified as a sensitivity range or interquartile range. These quantities answer different questions. Sampling intervals describe variation among sites or matched pairs. Rain peak-selection intervals include the choice among candidate peak times but remain conditional on one rainfall product and the fitted nuisance adjustment. The pH influence and XRF model ranges describe specific analysis choices and are not confidence intervals. The contamination interquartile range describes variation among linked profiles, not uncertainty in the median.

For paired contrasts and correlations, bootstrap resampling kept repeated observations from one site or aliquot pair together. For geographic R^2 , distance slopes and replacement shares, the delete-one-site jackknife removed the complete site; for pairwise distances it also removed every pair involving that site. Functional-profile bootstraps kept all profiles from a field site together. Exact machine-readable values, method labels, independent units and source files are in `analysis/v3/headline_uncertainty/headline_uncertainty.tsv`.

Table S12: Headline estimates and quantified uncertainty. Bootstrap intervals are percentile intervals. CI denotes a 95 % confidence interval.

Estimand	Estimate [interval or range]	Basis and independent unit
Geography		
Quadratic transect partial R^2	0.402 [0.325,0.478]	Site-jackknife CI; 60 sites
Aitchison slope per 100 km, surface	1.351 [0.905,1.796]	Site-jackknife CI; 60 sites
Aitchison slope per 100 km, shallow subsurface	1.510 [1.092,1.929]	Site-jackknife CI; 60 sites
Aitchison slope per 100 km, root-adjacent	1.180 [0.698,1.662]	Site-jackknife CI; 60 sites
Replacement share, surface	0.715 [0.636,0.793]	Site-jackknife CI; 60 sites
Replacement share, shallow subsurface	0.692 [0.607,0.776]	Site-jackknife CI; 60 sites
Replacement share, root-adjacent	0.747 [0.676,0.818]	Site-jackknife CI; 60 sites
Soil position		
Standardized composition, shallow minus surface	0.331 [0.306,0.404]	Paired-site bootstrap; 60 sites
Standardized composition, root-adjacent minus surface	0.433 [0.386,0.509]	Paired-site bootstrap; 60 sites
Standardized composition, root-adjacent minus shallow	0.478 [0.423,0.556]	Paired-site bootstrap; 60 sites

Continued on the next page

Table S12 continued

Estimand	Estimate [interval or range]	Basis and independent unit
Mean Shannon difference, shallow minus surface	0.075 [−0.089,0.235]	Paired-site bootstrap; 60 sites
Mean Shannon difference, root-adjacent minus surface	−0.130 [−0.321,0.079]	Paired-site bootstrap; 60 sites
Mean Shannon difference, root-adjacent minus shallow	−0.206 [−0.398,−0.001]	Paired-site bootstrap; 60 sites
Mean evenness difference, shallow minus surface	0.0125 [0.0008,0.0251]	Paired-site bootstrap; 60 sites
Mean evenness difference, root-adjacent minus surface	−0.0315 [−0.0448,−0.0186]	Paired-site bootstrap; 60 sites
Mean evenness difference, root-adjacent minus shallow	−0.0440 [−0.0572,−0.0313]	Paired-site bootstrap; 60 sites
Long-term climate		
Temperature versus Shannon, Spearman ρ	−0.711 [−0.833,−0.520]	Paired-site bootstrap; 60 sites
Temperature versus expected richness, ρ	−0.795 [−0.887,−0.639]	Paired-site bootstrap; 60 sites
Temperature versus evenness, ρ	−0.584 [−0.753,−0.359]	Paired-site bootstrap; 60 sites
Monthly rain versus Shannon, ρ	−0.470 [−0.671,−0.211]	Paired-site bootstrap; 60 sites
Monthly rain versus expected richness, ρ	−0.495 [−0.702,−0.226]	Paired-site bootstrap; 60 sites
Monthly rain versus evenness, ρ	−0.298 [−0.532,−0.023]	Paired-site bootstrap; 60 sites
Humidity versus Shannon, ρ	−0.701 [−0.834,−0.490]	Paired-site bootstrap; 60 sites
Humidity versus expected richness, ρ	−0.770 [−0.877,−0.593]	Paired-site bootstrap; 60 sites
Humidity versus evenness, ρ	−0.522 [−0.713,−0.266]	Paired-site bootstrap; 60 sites
pH and elemental composition		
Geographic partial R^2 before pH adjustment	0.358 [0.358,0.362]	Prespecified influence range; not a CI
Geographic partial R^2 after pH adjustment	0.196 [0.187,0.196]	Prespecified influence range; not a CI
pH transect partial R^2	0.846 [0.826,0.846]	Prespecified influence range; not a CI
Elemental-axis partial R^2	0.00358 [0.00311,0.00543]	Prespecified model range; not a CI
Recent rain		
NASA POWER richness change per mm at peak	179.6 [92.1,267.1]	Site-clustered sandwich CI; 60 sites
NASA POWER partial R^2 at peak	0.0364 [0.0060,0.0787]	Fixed-curve site bootstrap; 60 sites
NASA POWER selected peak, complete days	2.0 [1.5,4.5]	Peak-selection site bootstrap; 60 sites
Open-Meteo richness change per mm at peak	327.5 [141.1,514.0]	Site-clustered sandwich CI; 60 sites
Open-Meteo partial R^2 at peak	0.0384 [0.0057,0.0899]	Fixed-curve site bootstrap; 60 sites
Open-Meteo selected peak, complete days	1.0 [0.5,6.5]	Peak-selection site bootstrap; 60 sites
Predicted function		
PICRUSt2 pathway transect R^2	0.234 [0.155,0.314]	Site-jackknife CI; 60 sites
PICRUSt2 position, shallow minus surface	0.330 [0.296,0.501]	Paired-site bootstrap; 60 sites
PICRUSt2 position, root-adjacent minus surface	0.429 [0.350,0.566]	Paired-site bootstrap; 60 sites
PICRUSt2 position, root-adjacent minus shallow	0.453 [0.412,0.549]	Paired-site bootstrap; 60 sites
Median within-sample PICRUSt2-shotgun ρ	0.743 [0.731,0.751]	Whole-site bootstrap; 125 profiles
Community-mean PICRUSt2-shotgun ρ	0.806 [0.793,0.812]	Whole-site bootstrap; 125 profiles
Assay controls		
PMA richness difference, treated minus untreated	−163.3 [−274.0,−69.8]	Paired-aliquot bootstrap; nine pairs
PMA Shannon difference, treated minus untreated	0.130 [−0.154,0.467]	Paired-aliquot bootstrap; nine pairs
Candidate-read fraction per linked profile	0.00403 [0.00155,0.00987]	Empirical median [IQR]; 217 profiles

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